

Steel Defect Detection - Project Files Report

- **src/models.py:** Defines the AI models (U-Net, FPN, DeepLabV3+).
- **src/dataset.py:** Loads images and mask data for training.
- **src/config.py:** Stores hyperparameters (batch size, learning rate).
- **src/utils.py:** Helper functions for metrics and encoding.
- **train.py:** Main script to train the models.
- **predict.py:** Runs inference on local images.
- **api/app.py:** FastAPI backend server for the web app.
- **static/:** Frontend UI files (HTML, CSS, JS).
- **app.py:** Gradio app for Hugging Face Spaces free-tier.
- **notebooks/02_live_demo_colab.ipynb:** Public Colab web demo.
- **space_static/:** Static landing page deployed to Hugging Face.
- **Dockerfile:** Docker container configuration.
- **deploy.py:** Automation script for pushing code and models.